

Worksheet: Material balances on a CSTR

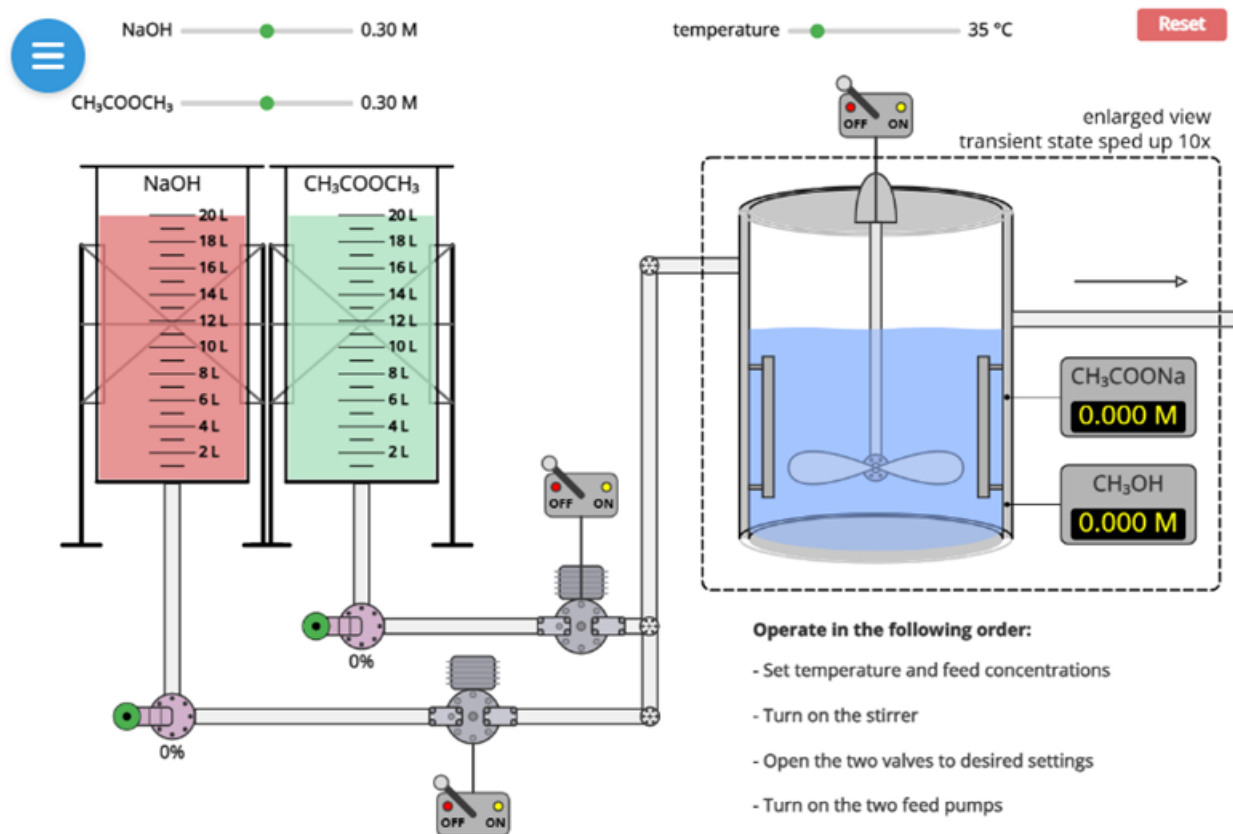
Name(s) _____

In this experiment, two liquid reactants in aqueous solutions flow into a steady-state, continuous stirred tank reactor (CSTR). The objective is to determine the rate constant and the reaction order with respect to one of the reactants.

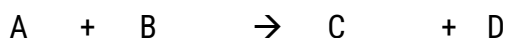
Student learning objectives

1. Apply steady-state mass balances to a chemical reactor.
2. Explain how changing the flow rate to a CSTR changes the rate of reaction and the conversion.

Experimental Diagram



The reaction is: $\text{NaOH} + \text{CH}_3\text{COOCH}_3 \rightarrow \text{CH}_3\text{COONa} + \text{CH}_3\text{OH}$



sodium hydroxide + methyl acetate $\rightarrow$ sodium acetate + methanol

Assumptions

The reactor is well mixed, so the concentrations of reactants and products in the reactor are identical to those concentrations leaving the reactor.

Because the reactor is well mixed, the temperature is the same everywhere in the reactor.

The density of the liquid feeds is the same as the density of the effluent from the reactor so the volumetric flow rate into the reactor is the same as the volumetric flow rate leaving the reactor.

Questions to answer before beginning the experiment

1. If the volumetric flow rate to an isothermal CSTR increases, does the reactant concentration in the reactor increase or decrease for a first- or second-order reaction? Why?
2. If the volumetric flow rate increases to an isothermal CSTR, is more product produced per hour or is less product produced per hour? Explain.

Notation

v_A = volumetric feed rate from the tank containing reactant A (L/s)

v_B = volumetric feed rate from the tank containing reactant B (L/s)

$v = v_A + v_B$ = total volumetric flow rate into reactor (L/s)

$C_{A,feed}$ = concentration of reactant A in the feed tank (mol/L)

$C_{B,feed}$ = concentration of reactant B in the feed tank (mol/L)

C_{A0} = concentration of reactant A entering the reactor (mol/L)

C_{B0} = concentration of reactant B entering the reactor (mol/L)

C_A = concentration of reactant A in the reactor (mol/L)

C_B = concentration of reactant B in the reactor (mol/L)

C_C = concentration of product C in the reactor (mol/L)

C_D = concentration of product D in the reactor (mol/L)

t_A = elapsed time for feed tank A volume change (s)

t_B = elapsed time for feed tank B volume change (s)

ΔV_A = feed tank A volume change in time t_A (L)

ΔV_B = feed tank B volume change in time t_B (L)

k_1 = first-order rate constant (s^{-1})

k_2 = second-order rate constant ($L\ mol^{-1}s^{-1}$)

V = reactor volume (1.0 L)

τ = reactor residence time (V/v)

X_A = conversion of reactant A

Experiment #1

The reactor, which has a volume of 1.0 L, starts filled with solvent (water). Note that in the simulation the size of the reactor is enlarged relative to the size of the feed tanks.

1. Select a temperature _____
2. Move the stirrer switch to ON.
3. Select concentrations of the two feed tanks ($C_{A,feed}$, $C_{B,feed}$) and record the values in Table 1.
4. Move the valve openings to a percentage greater than zero for the NaOH and the $\text{CH}_3\text{COOCH}_3$ solution feeds.
5. Move both pump switches to ON to start NaOH and methyl acetate flow into the reactor. Note that the pump switches cannot be moved to ON unless the stirrer is ON.
6. Allow time for the system to reach steady state; this time is typically about four times the reactor residence time τ , which is V/v . In the simulation, the time to reach steady state is $1/10^{\text{th}}$ of the time for a physical reactor.
7. Use the stopwatch on your phone to measure the volume changes (ΔV_A , ΔV_B) of the two feed tanks for elapsed times (t_A , t_B) and record in Table 1. Calculate the volumetric flow rates leaving the two feed tanks (v_A , v_B) and the total volumetric flow rate (v) into the reactor and record the values in Table 1.
8. Because mixing the two feed streams dilutes the reactant concentrations, the concentrations entering the reactor (C_{A0} , C_{B0}) not the same as the concentrations in the feed tanks ($C_{A,feed}$, $C_{B,feed}$). Calculate their concentrations entering the reactor (C_{A0} , C_{B0}) and enter the values in Table 1:

$$C_{A0} = \frac{v_A C_{A,feed}}{v_A + v_B} \quad C_{B0} = \frac{v_B C_{B,feed}}{v_A + v_B}$$

Table 1 Reactor feed conditions

Exp #	$C_{A,feed}$	$C_{B,feed}$	t_A	ΔV_A	v_A	t_B	ΔV_B	v_B	v	C_{A0}	C_{B0}
1											
2											
3											
4											
5											
6											
7											
8											

- Record C_{A0} and the product concentrations (CH_3COONa , C_C ; CH_3OH , C_D) in the reactor in Table 2.
- Use stoichiometry to calculate the reactant concentrations in the reactor and record those values in Table 2.

Table 2 Reactor concentrations and reactant A conversion

Exp #	C_{A0}	C_A	C_B	C_C	C_D	X_A
1						
2						
3						
4						
5						
6						
7						
8						

- Calculate the conversion of NaOH (reactant A) and record the value in Table 2. Conversion of A (X_A) for a constant-density system is:

$$X_A = \frac{C_{A0} - C_A}{C_{A0}}$$

- Make measurements at 8 conditions and record values in Tables 1 and 2.
- For measurements using the same feed tank concentrations but different valve openings, change the valve openings, repeat steps 6 - 10. For measurements at different feed tank concentrations, click the "Reset" button, change the feed tank concentrations, repeat steps 2- 10. Make measurements at different conditions to determine the order of the reaction (n) with respect to $\text{CH}_3\text{COOCH}_3$. Assume the reaction is first order in NaOH (reactant A).

$$\text{rate} = v(C_{A0} - C_A) = kC_A C_B^n V$$

Reaction order for $\text{CH}_3\text{COOCH}_3$ _____

Data Analysis

- After the order of reaction is determined with respect to $\text{CH}_3\text{COOCH}_3$, use the data in the tables to obtain the rate constant at each condition using one of these CSTR mass balances below. Record the values in Table 3 and calculate an average.

mass balance for reactant A (NaOH) for reaction that is first-order reaction in NaOH and zero order with respect to $\text{CH}_3\text{COOCH}_3$ is:

Accumulation = input - output + generation

$$0 = vC_{A0} - vC_A - k_1 C_A V$$

mass balance for reactant A for a reaction that is first order in NaOH and first order in $\text{CH}_3\text{COOCH}_3$ (second-order overall) is:

$$0 = vC_{A0} - vC_A - k_2C_AC_BV$$

2. Since the reaction is $A + B \rightarrow C + D$, stoichiometry requires that the amount of A that reacts must equal the amount of B that reacts: $C_{A0} - C_A = C_{B0} - C_B$

Thus, the mass balance in terms of C_A for a second-order reaction is:

$$0 = vC_{A0} - vC_A - k_2C_A(C_{B0} - C_{A0} + C_A)V$$

Table 3 Rate constants

Experiment	Rate constant
1	
2	
3	
4	
5	
6	
7	
8	
Average	

Experiment #2

Select a temperature of 30°C, select reactant feed concentrations and keep these concentrations for these experiments. Make measurements for four flow rates that cover a wide range of residence times. For each flow rate, the valve openings should be the same for each reactant. Record the values in Tables 4 and 5. Plot NaOH conversion versus residence time (τ).

Table 4 Reactor feed conditions

Exp #	$C_{A,feed}$	$C_{B,feed}$	t_A	ΔV_A	v_A	t_B	ΔV_B	v_B	v	C_{A0}	C_{B0}
1											
2											
3											
4											

Table 5 Reactor concentrations and reactant A conversion

Exp #	C_{A0}	C_A	C_B	C_C	C_D	X_A
1						
2						
3						
4						

Questions to answer

1. Explain the reason for the trend observed in the NaOH conversion versus residence time plot.
2. If this experiment were conducted in the laboratory, what might cause the conversion in the reactor to differ from that expected based on the known value of the rate constant?
3. The reactor is assumed to be isothermal. In a real reactor, what would be needed to keep the reactor temperature constant? Explain.
4. Why can the total amount of products produced per time increase even when conversion decreases?
5. Why is the reactor outlet concentration equal to the reactor concentration for a well-mixed CSTR?
6. What safety measures would you employ if making this measurement in the laboratory?